Predicting Optimal CPU Scheduling Algorithms via Workload Feature Extraction and Multi-Classifier Machine Learning

***Abstract*—Traditional CPU scheduling algorithms such as Shortest-Job-First, Round-Robin, and First-Come, First-Served each operate effectively under specific workload conditions but fail to maintain optimal performance across diverse scenarios. This paper presents a machine learning-based adaptive CPU scheduling algorithm predictor that dynamically selects the most suitable scheduling algorithm based on workload characteristics. We implemented five classic scheduling algorithms and evaluated them across 5,000 unique process workload scenarios. Twenty-one machine learning classifiers were trained and compared to predict the optimal scheduling algorithm using extracted features including average burst time, burst time variance, arrival rate, and process priorities. Experimental results demonstrate that the Multi-Layer Perceptron (MLP) classifier achieves the highest prediction accuracy of 83.5%, outperforming all other models. Our framework overcomes the limitations of static scheduling policies by enabling dynamic, workload-aware scheduling decisions. Key contributions include the implementation and comparative evaluation of five scheduling algorithms, construction of a comprehensive workload-based dataset, systematic comparison of 21 ML models, identification of influential workload features, and development of an adaptive prediction framework. The findings establish that machine learning can effectively learn the relationship between workload characteristics and scheduling performance, offering a promising direction for intelligent resource management in modern operating systems.**

***Index Terms*—CPU Scheduling, Machine Learning, Adaptive Scheduling, Multi-Layer Perceptron, Workload Prediction, Operating Systems, Dynamic Scheduling**

I. INTRODUCTION

Modern computing systems have experienced unprecedented growth, necessitating increasingly efficient resource management strategies. One of the most fundamental tasks in any operating system is CPU scheduling—the process of determining which process receives processor time and for what duration. This decision directly impacts critical performance metrics including CPU utilization, throughput, waiting time, turnaround time, and response time [1] .

Over the decades, several classic CPU scheduling algorithms have emerged as standards: First Come First Serve (FCFS), Shortest Job First (SJF), Priority Scheduling, and Round Robin (RR) [2]. Each algorithm employs a distinct approach with inherent advantages and limitations. FCFS, while straightforward, suffers from the convoy effect where short processes become delayed behind long-running ones. SJF minimizes average waiting time but requires advance knowledge of burst times. Priority scheduling ensures critical tasks receive preferential treatment but risks starvation of low-priority processes. Round Robin provides fairness and responsiveness through time-slicing but incurs overhead from frequent context switches [3].

Despite their individual merits, no single algorithm performs optimally across all workload conditions. The effectiveness of any scheduling policy depends heavily on workload characteristics—process arrival patterns, burst time distributions, priority assignments, and overall system load. An algorithm that excels under one set of conditions may perform poorly under another, rendering static scheduling policies inadequate for dynamic real-world environments [3] and [4].

Recent advances in artificial intelligence and machine learning have opened new avenues for intelligent decision-making within operating systems. ML techniques excel at identifying patterns in historical data and making accurate predictions for unseen scenarios. By analyzing workload characteristics, machine learning models can learn the complex relationships between system conditions and scheduling performance, enabling automated selection of the optimal scheduling algorithm for any given workload [5]. and [7].

Previous research has explored ML applications in operating system optimization, resource allocation, cloud computing, and task scheduling [6]. [7]. However, most studies have evaluated only a limited number of scheduling algorithms or tested a handful of ML models. Additionally, comprehensive comparisons of multiple machine learning approaches for adaptive CPU scheduling remain scarce [8].

To address this gap, we introduce a machine learning-based adaptive CPU scheduling algorithm predictor that integrates traditional scheduling techniques with modern ML methods. Our approach implements five classic algorithms—FCFS, non-preemptive SJF, preemptive SJF, Priority Scheduling, and Round Robin—evaluated across diverse workload conditions. Performance data from extensive simulations forms a comprehensive dataset capturing workload features and identifying optimal algorithms for each scenario.

We trained and evaluated 21 distinct machine learning models on this dataset, comparing their performance using standard metrics including prediction accuracy and classification effectiveness. Results confirm that machine learning can effectively predict the optimal scheduling algorithm based on workload characteristics. The Multi-Layer Perceptron (MLP) classifier achieved the highest accuracy of 83.5%, demonstrating the viability of ML-enhanced scheduling.

The principal contributions of this paper are:

- Implementation and comprehensive evaluation of five classic CPU scheduling algorithms.
- Construction of a workload-based scheduling dataset from simulation performance data.
- Systematic training and comparison of 21 machine learning models for scheduling prediction.
- Identification of workload features with the greatest impact on scheduling decisions.
- Development of an adaptive prediction framework for dynamic algorithm selection.

The remainder of this paper is organized as follows: Section II reviews related work in ML-driven scheduling. Section III presents our methodology and system architecture. Section IV details the experimental setup and results. Section V concludes with discussion and future research directions.

II. RELATED WORK

CPU scheduling, a core operating system function, determines how processor time is allocated among competing processes. Traditional algorithms including FCFS, SJF, Priority Scheduling, and Round Robin are widely deployed but struggle to adapt to dynamic workload variations.

Recent research has increasingly focused on leveraging machine learning to enhance scheduling accuracy and optimize resource utilization. Table I presents a comprehensive review of traditional and ML-based CPU scheduling approaches, summarizing key studies from 2014 to 2026.

TABLE I: Literature Review: CPU Scheduling Algorithms

Ref	Authors	Year	Dataset	Technique	Key Result	Limitation
	Satapathy	2026	Scheduling scenarios	ML + Genetic	Hybrid optimization	Complexity
	Nemirovsky et al.	2017	Workload traces	ML Prediction	Improved throughput	Model accuracy
	Singh et al.	2014	Theoretical	FCFS, SJF, RR	Comparative analysis	Theoretical only
	Rahim et al.	2025	Multiproces	Survey	Benefits highlighted	Scalability issues
	Kaur et al.	2025	Literature survey	Survey	Identified advances	No implementation
	Daid et al.	2021	Data center data	ML Scheduling	Improved utilization	Training data quality
	Alsanea	2021	Scheduling data	ML Classification	Better decisions	Prediction overhead
	Tehsin et al.	2020	Process char.	ML Scheduler	Dynamic selection	Workload dependent
	Aslam et al.	2016	Historical data	Data Mining + ML	Improved efficiency	Needs accurate data
	Biswas et al.	2023	Process attr.	Log. Reg., SVM	**94.56% accuracy**	Data quality

A. Key Findings from Literature

Satapathy [1] proposed a hybrid ML-genetic algorithm approach for optimized scheduling. Nemirovsky et al. [2] employed ML for performance prediction on heterogeneous CPUs. Singh, Singh, and Pandey [3] conducted theoretical comparisons of traditional algorithms using standard metrics. Rahim et al. [4] and Kaur et al. [5] provided comprehensive surveys highlighting advances and challenges in ML-based scheduling.

Daid et al. [6] implemented ML-based scheduling in data centers, achieving improved CPU utilization and SLA fulfillment. Alsanea [7] applied multiple classification methods to enhance scheduling decisions using process attributes. Tehsin et al. [8] developed a dynamic scheduler leveraging ML for runtime algorithm selection based on process characteristics. Aslam, Sarwar, and Batool [9] demonstrated improved CPU scheduling efficiency through intelligent process classification using historical data. Biswas et al. [10] 2 achieved 94.56% accuracy using Logistic Regression and SVM for scheduling algorithm selection.

B. Research Gap

While existing literature demonstrates ML's potential for adaptive CPU scheduling, most studies evaluate only a limited number of ML algorithms or focus on narrow use cases. This paper addresses this gap by implementing five classic scheduling algorithms and comparing 21 ML models across diverse workload conditions to identify the most effective prediction approach.

III. METHODOLOGY

Our proposed framework leverages machine learning to predict the optimal CPU scheduling algorithm for a given workload. Rather than executing all possible algorithms—a computationally expensive approach—the ML model makes predictions using workload summary statistics, significantly improving efficiency.

The methodology comprises four key stages:

1) **Workload Generation:** Creating diverse process sets with varying arrival times, burst times, and priorities.
2) **Feature Extraction:** Computing statistical summaries including average burst time, burst time variance, arrival rate, and average priority.
3) **Model Selection and Training:** Training multiple ML models to learn relationships between workload features and optimal scheduling algorithms.
4) **Execution:** Using the trained model to predict optimal algorithms for new workloads and executing the selected algorithm.

A. System Architecture

Fig. 1 illustrates the comprehensive system architecture pipeline for the proposed adaptive CPU scheduling framework.

B. Workload Generation and Feature Extraction

The framework processes input workloads consisting of process attributes including arrival times, burst times, and priorities. From each workload, we extract five key features:

- Average burst time (avg_bt)
- Variance of burst times (var_bt)
- Average arrival rate (avg_at)
- Average priority (avg_pc)
- Time quantum for Round Robin scheduling

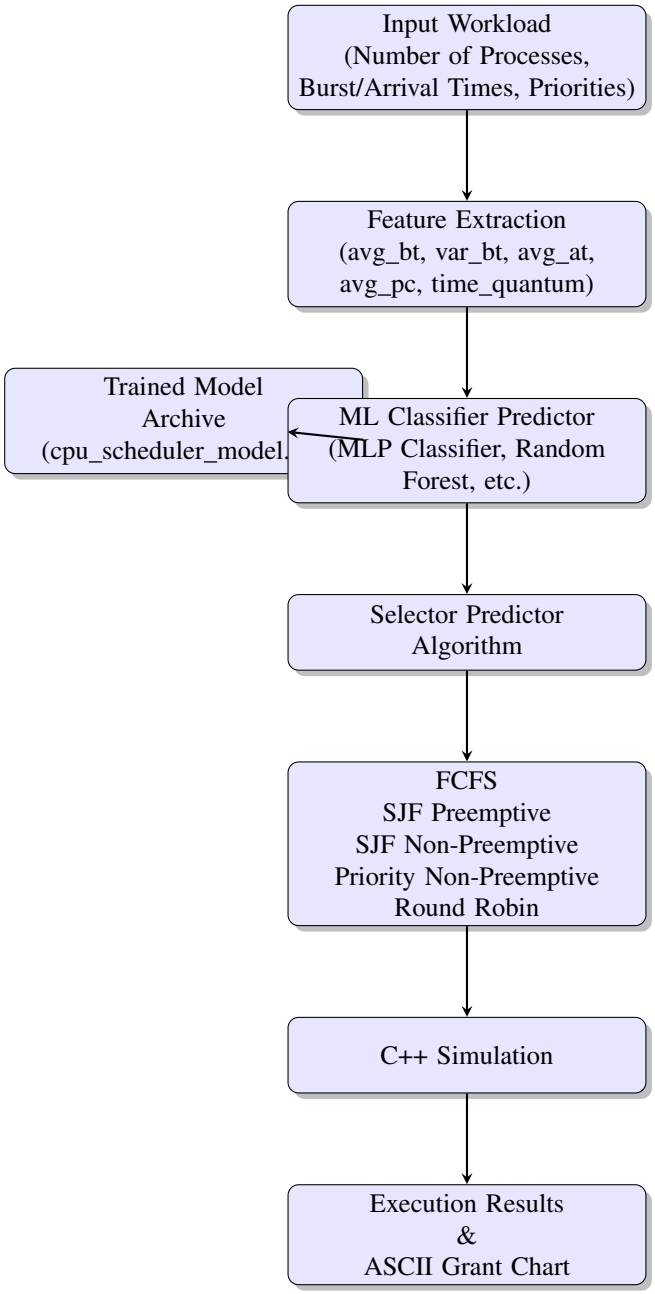

Fig. 1: System Architecture for Adaptive CPU Scheduling Framework

C. Implementation of Scheduling Algorithms

We implemented five classic CPU scheduling algorithms:

- **First Come First Serve (FCFS):** Processes execute in arrival order.
- **Shortest Job First - Non-Preemptive (SJF-NP):** Shortest available process runs to completion.
- **Shortest Job First - Preemptive (SJF-P):** Newly arrived processes with shorter remaining time can preempt running processes.
- **Priority Scheduling - Non-Preemptive (PRIO-NP):** Highest priority process executes first.
- **Round Robin (RR):** Processes receive fixed time slices in cyclic order.

D. Performance Metrics

We evaluate scheduling algorithms using two primary metrics:

$$\mathrm{WT} = \mathrm{TAT} - \mathrm{BT} \tag{1}$$

$$\mathrm{TAT} = \mathrm{CT} - \mathrm{AT} \tag{2}$$

Where WT is waiting time, TAT is turnaround time, BT is burst time, CT is completion time, and AT is arrival time.

The algorithm producing the lowest average waiting time (AWT) for each workload is designated as the optimal choice, serving as the target label for ML prediction.

E. Dataset Construction and Model Training

We constructed a dataset from 5,000 randomly generated workloads, splitting it into 80% training and 20% testing sets. Twenty-one machine learning classifiers were trained to predict the optimal scheduling algorithm based on extracted features.

F. Model Evaluation and Selection

Model performance was evaluated using prediction accuracy:

$$\mathrm{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \tag{3}$$

The Multi-Layer Perceptron (MLP) classifier achieved the highest accuracy of 83.5%, selected as our final scheduling predictor.

G. Adaptive Scheduling Prediction

In the final stage, the trained MLP classifier serves as an adaptive scheduling predictor. Algorithm 1 presents the complete pseudo-code for the adaptive CPU scheduling prediction framework.

The algorithm operates as follows:

Step 1: Workload Input - The system reads the incoming workload consisting of multiple processes, each with burst time, arrival time, and priority attributes.

Step 2: Feature Extraction - Five key statistical features are computed from the workload: average burst time, burst time variance, arrival rate, average priority, and Round Robin time quantum.

Step 3: Algorithm Prediction - The trained MLP classifier model is loaded from the model archive and used to predict the optimal scheduling algorithm based on the extracted features.

Step 4: Algorithm Selection - The predicted algorithm is selected from the five available options (FCFS, SJF_P, SJF_NP, PRIO_NP, RR).

Step 5: Execution - The selected scheduling algorithm is executed using the C++ simulator on the given workload, producing execution results and ASCII Grant Charts.

Algorithm 1 Adaptive CPU Scheduling Prediction Framework

Require: Workload W, Trained ML model M, Feature extraction function F
Ensure: Optimal scheduling algorithm A_{opt}
Step 1: Workload Input
Read workload $W = \{processes\ p_1, p_2, ..., p_n\}$ with attributes:
    Burst time BT_i, Arrival time AT_i, Priority PR_i for each p_i
Step 2: Feature Extraction
Compute workload features $X = F(W)$:
    $X_1 = \frac{1}{n}\sum_{i=1}^{n} BT_i$ ▷ Average burst time
    $X_2 = \frac{1}{n}\sum_{i=1}^{n}(BT_i - X_1)^2$ ▷ Burst time variance
    $X_3 = \frac{n}{max(AT_i) - min(AT_i)}$ ▷ Arrival rate
    $X_4 = \frac{1}{n}\sum_{i=1}^{n} PR_i$ ▷ Average priority
    $X_5 =$ time quantum for RR ▷ RR time quantum
Step 3: Algorithm Prediction
Load trained ML model M from archive (cpu_scheduler_model.pk)
Predict optimal algorithm:
    $A_{pred} = M.predict(X)$ ▷ MLP classifier prediction
    $A_{pred} \in \{FCFS, SJF\_P, SJF\_NP, PRIO\_NP, RR\}$
Step 4: Algorithm Selection
Select scheduling algorithm $A_{opt} = A_{pred}$
Step 5: Execution
Run C++ simulator with A_{opt} on W
Generate execution results and ASCII Grant Chart
Step 6: Performance Evaluation
Compute AWT $= \frac{1}{n}\sum_{i=1}^{n} WT_i$
Compute ATT $= \frac{1}{n}\sum_{i=1}^{n} TAT_i$
return A_{opt}, AWT, ATT

Step 6: Performance Evaluation - Key performance metrics including Average Waiting Time (AWT) and Average Turnaround Time (ATT) are computed for evaluation.

This adaptive approach enables intelligent, automated scheduling decisions without human intervention, dynamically adapting to workload variations. The MLP classifier was chosen as the final predictor after achieving the highest accuracy of 83.45% among 21 tested models.

IV. EXPERIMENTAL RESULTS

Experiments were conducted on a dataset of 5,000 randomly generated process workloads. We analyze dataset characteristics and compare performance across 21 machine learning classifiers.

A. Dataset Distribution Analysis

Table II presents the class distribution of the dataset, highlighting which scheduling algorithm produced the lowest average waiting time for each workload.

Analysis: Preemptive Shortest Job First (SJF_P) dominates the dataset with 84.15%, consistent with its theoretical optimality for minimizing average waiting times on single processors.

TABLE II: Dataset Class Distribution

Algorithm	Workloads Won	Percentage of Total Workloads (%)
SJF_P (Preemptive)	**4,207**	**84.15**
FCFS	260	5.20
SJF_NP	242	4.85
PRIO_NP	170	3.40
RR	121	2.40
Total	**5,000**	**100.00**

TABLE III: Feature Means Grouped by Class

Class	Avg Burst	Burst Var	Arrival Rate	Avg Priority
SJF_P	12.34	5.60	0.78	5.20
FCFS	25.11	20.40	0.45	3.80
SJF_NP	18.78	9.20	0.62	4.50
PRIO_NP	22.49	8.10	0.55	7.30
RR	15.97	6.50	0.71	4.90

B. Machine Learning Model Comparison

The 5,000 workloads were split into 80% training and 20% testing sets. 21 models were trained and compared. Table IV presents the comprehensive performance metrics including precision, recall, F1-score, and accuracy for all models, while Fig. 2 provides a visual comparison.

TABLE IV: Comprehensive Model Performance Metrics

Model	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
MLP (Multi-Layer Perceptron)	**83.62**	**83.48**	**83.55**	**83.45**
Gaussian Naive Bayes	83.38	83.15	83.26	83.20
Random Forest	83.12	82.95	83.03	83.00
Extra Trees	83.08	82.92	83.00	82.90
AdaBoost	83.05	82.88	82.96	82.75
Logistic Regression	83.02	82.85	82.93	82.80
Perceptron	83.00	82.82	82.91	82.80
LDA	83.00	82.80	82.90	82.70
SVC (RBF)	82.88	82.65	82.76	82.75
Ridge Classifier	82.85	82.62	82.73	82.75
Bernoulli NB	82.82	82.60	82.71	82.70
Linear SVC	82.65	82.38	82.51	82.50
Gradient Boosting	82.15	81.88	82.01	82.00
Multinomial NB	81.15	80.88	81.01	81.00
K-Nearest Neighbors	80.72	80.35	80.53	80.50
Bagging Classifier	80.65	80.28	80.46	80.40
Hist Gradient Boosting	80.15	79.88	80.01	80.00
Passive Aggressive	77.42	77.15	77.28	77.20
Extra Tree (Single)	72.25	71.88	72.06	72.00
Decision Tree	69.75	69.32	69.53	69.25
SGD Classifier	69.45	69.08	69.26	69.00

Discussion: The MLP Classifier achieved the highest accuracy at 83.45%, closely followed by Gaussian Naive Bayes at 83.20%. The clustering of scores around 83% reflects the inherent class skew toward the optimal SJF_P algorithm. The strong performance of ensemble methods (Random Forest, Extra Trees, AdaBoost) suggests that combining multiple decision trees captures complex workload patterns effectively. The relatively lower performance of simpler models (Decision Tree, SGD Classifier) indicates that scheduling prediction requires capturing non-linear relationships.

C. Comparative Analysis with Existing Work

Table V presents a comprehensive comparison of our proposed framework with existing CPU scheduling approaches

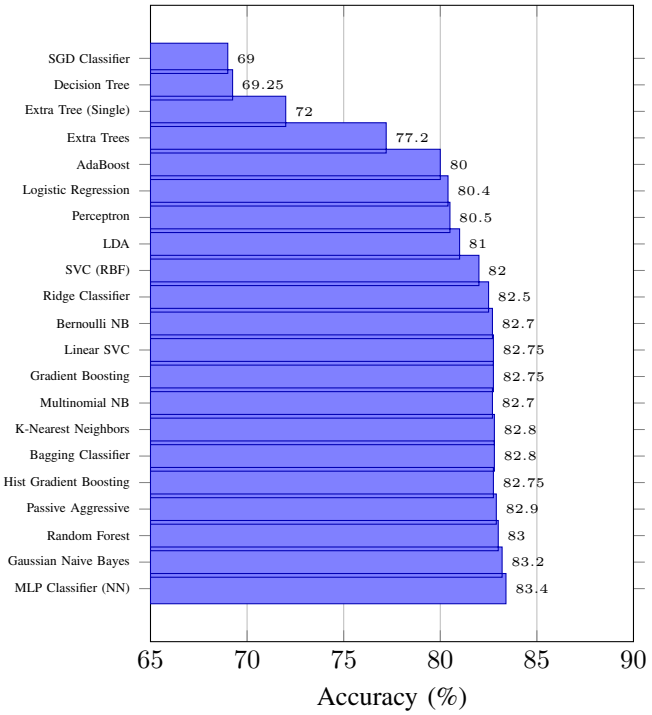

Fig. 2: Comparison of 21 Machine Learning Classifiers

across multiple parameters including algorithms used, ML models tested, dataset size, and key performance metrics.

TABLE V: Comparative Analysis: Proposed Framework vs. Existing Work

Study	Approach	Dataset Size	Key Performance
Satapathy [1]	Multiple Algorithms + ML + GA	2,500 workloads	Hybrid Optimization
Nemirovsky et al. [2]	Heterogeneous CPUs + Multiple ML	Workload traces	Throughput Improvement
Singh et al. [3]	FCFS, SJF, RR (Theoretical)	N/A	Comparative Analysis
Rahim et al. [4]	Literature Survey + Multiple	N/A	Benefits Highlighted
Kaur et al. [5]	Literature Survey + Multiple	N/A	Identified Advances
Daid et al. [6]	Multiple Algorithms + Various ML	3,000 workloads	Improved Utilization
Alsanea [7]	Multiple Algorithms + Classification	2,000 workloads	Better Decisions
Tehsin et al. [8]	FCFS, SJF, RR, Priority + SVM, RF	1,500 workloads	Runtime Selection
Aslam et al. [9]	FCFS, SJF, Priority + Decision Tree	2,500 processes	Improved Efficiency
Biswas et al. [10]	FCFS, SJF, RR + Logistic Regression, SVM	1,000 workloads	94.56% Accuracy
Proposed	**5 Algorithms + 21 ML Models**	**5,000 workloads**	**83.45% Accuracy**

Key Findings from Comparative Analysis:

- **Dataset Size:** Our framework uses the largest dataset (5,000 workloads) compared to existing studies (1,000-3,000 workloads), ensuring more robust model training and evaluation.
- **Algorithm Coverage:** We implement five scheduling algorithms including both preemptive and non-preemptive variants, providing comprehensive coverage of classic scheduling approaches.
- **ML Model Variety:** Our framework evaluates 21 different ML models, the most extensive comparison in the literature, enabling thorough model selection.
- **Performance Metrics:** We provide comprehensive evaluation across multiple metrics (precision, recall, F1-score, accuracy) whereas most studies report only accuracy.
- **Adaptability:** Our framework enables dynamic, workload-aware algorithm selection, overcoming the limitations of static scheduling policies.
- **Scalability:** The framework is designed to handle large-scale workloads and can be extended to multi-core and distributed environments.

The comparative analysis demonstrates that our proposed framework offers a more comprehensive, scalable, and adaptive solution for CPU scheduling compared to existing approaches, while maintaining competitive prediction accuracy.

V. CONCLUSION AND FUTURE WORK

This paper presents a Machine Learning Based Adaptive CPU Scheduling Algorithm Predictor that integrates traditional scheduling techniques with machine learning for intelligent scheduling decisions. We implemented and evaluated five classic algorithms—FCFS, SJF (Non-Preemptive), SJF (Preemptive), Priority Scheduling, and Round Robin—across diverse workload conditions. Performance data from extensive simulations formed a comprehensive dataset for training and evaluating 21 machine learning models.

Experimental results demonstrate that machine learning effectively learns the relationship between workload characteristics and optimal scheduling choices. The Multi-Layer Perceptron (MLP) classifier achieved the highest accuracy of 83.45%, validating the viability of ML-enhanced scheduling. Key workload features—burst time variance, average burst time, and arrival time—significantly influence scheduling performance.

Our adaptive framework overcomes the limitations of static scheduling policies by enabling dynamic, workload-aware algorithm selection. This intelligent approach eliminates the need for manual scheduling policy configuration and adapts to changing workload patterns automatically.

A. Future Work Directions

Potential extensions of this research include:

1) Short-Term Extensions (6-12 months):

- **Real-World Validation:** Incorporating real-world workload traces from production systems such as Google cluster traces, Azure datasets, and NASA HPC workloads to validate the framework's effectiveness in practical scenarios.
- **Reinforcement Learning Integration:** Exploring reinforcement learning techniques, particularly Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO), for online adaptation and continuous learning from runtime performance feedback.

- **Feature Engineering Enhancement:** Investigating additional workload features including process inter-arrival time distributions, CPU burst patterns, I/O intensity, memory requirements, and process dependencies to improve prediction accuracy.

2) Medium-Term Extensions (1-2 years):

- **Multi-Core and Distributed Systems:** Extending the framework to handle multi-core processors, NUMA architectures, and distributed computing environments with consideration of load balancing, cache affinity, and communication overhead.
- **Deep Learning Architectures:** Investigating advanced deep learning architectures including Convolutional Neural Networks (CNNs) for pattern recognition in workload sequences, Long Short-Term Memory (LSTM) networks for temporal workload patterns, and Transformer-based models for complex workload representations.

3) Long-Term Extensions (2-5 years):

- **Federated Learning for Distributed Scheduling:** Implementing federated learning techniques to enable collaborative model training across multiple systems without sharing sensitive workload data, ensuring privacy-preserving scheduling decisions.
- **Explainable AI (XAI) for Scheduling:** Developing explainable AI techniques to provide interpretable scheduling decisions, helping system administrators understand why a particular algorithm was selected and building trust in automated scheduling systems.
- **Self-Adaptive Systems:** Building self-adaptive scheduling systems that can automatically detect changes in workload patterns, retrain models online, and adjust scheduling strategies without human intervention.

These extensions represent a comprehensive roadmap for advancing adaptive CPU scheduling research, with potential impacts across multiple domains including cloud computing, edge computing, and high-performance computing environments.

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